

TURNING POINT

SELF-CONTAINED CARTRIDGES

In the early 19th century, the discovery of chemical primers and the invention of percussion ignition led to an even greater advance. It became possible to combine the key elements required for a gun to fire—primer, propellant, and projectile—into a single unit, the self-contained, or unitary, cartridge. Following a period of experimentation, the solid-drawn, center-fire metallic cartridge evolved in the 1870s, triggering a new era in firearms technology. The subsequent development of repeating rifles, self-loading pistols, and machine-guns ultimately culminated in the weapons seen today.



▲ METALLIC CARTRIDGE

All metallic cartridges, such as this .44in-40 Winchester cartridge, contain three main elements within a metal shell. These are a propellant (gunpowder), projectile (bullet), and chemical primer.

Although the percussion cap containing chemical primer (see pp.80–81) made muzzle-loaders far more reliable, inserting gunpowder and ball separately down the muzzle, and then adding a primer, was a laborious process. Early attempts to unite a breech-loading system with percussion-cap ignition resulted in the creation of some breech-loading guns in the mid-19th century. These guns suffered from the problem of leakage of gas at the breech because the paper

or linen cartridge used did not form a gas-tight seal. However, the door to successful breech-loading guns had already been opened in the early 19th century with the invention of a “self-contained” cartridge.

UNITARY CARTRIDGES

Patented by gunsmith Jean Pauly in France in 1812, the first self-contained cartridge had a paper casing and a metal base. It worked perfectly in careful hands but it was not rugged enough for military use. In the following years, the cartridge was reinvented in several ways to improve the ruggedness, the ease of loading, and ignition, and the gas seal. Pauly's ex-employee Casimir Lefauchaux created a “pin-fire” cartridge of cardboard and brass in 1836, in which a metal pin struck and ignited the chemical primer in the cartridge. In 1841, another of Pauly's former employees, Nikolaus von Dreyse, created a cartridge with a combustible paper case. They both worked and had limited success, but they had too many drawbacks for widespread adoption.

In 1846, Parisian gunsmith Benjamin Houllier took a major step by creating a cartridge case pressed from a disc of copper or brass. Its all-metal, single-piece design properly sealed the breech. American Benjamin Tyler Henry used the same construction, but added a hollow rim filled with chemical primer, creating the first rim-fire cartridge in 1860.

BEGINNINGS OF CENTER-FIRE

Rim-fire cartridges had to be handled carefully, because they were liable to accidental discharge and the rim could burst in use. A major breakthrough—the center-fire cartridge—held the chemical primer in a percussion cap fixed in the center of the cartridge's base. Designed in Britain by Colonel Boxer, the cartridge did not need to be aligned while loading, as with

pin-fires, and could be reloaded easily, unlike rim-fires. However, it had a complex composite case. US inventor Hiram Berdan developed a one-piece brass case, which was to become the standard for most cartridges in the future. By the late 1870s, center-fire metallic cartridges, similar to today's, had taken hold.

» BEFORE

Before the advent of the self-contained cartridge, the loading of a gun required a user to place the correct charge of propellant in the barrel, along with a projectile and some wadding to hold the propellant and projectile in place, in the correct sequence. Next, he had to employ an external means of ignition, as there was no primer inside the barrel.



EARLY PAPER CARTRIDGE

- **SINGLE-SHOT WEAPONS** were the norm of the day.

- **PAPER CARTRIDGES** contained the correct charge of gunpowder and a projectile. They needed to be torn open before loading a gun.

- **AN INCORRECT LOADING SEQUENCE** would leave the gun useless until it could be unloaded and then reloaded correctly.

- **AN INCORRECTLY RAMMED PROJECTILE**, one not placed firmly on top of the gunpowder, could cause the gun barrel to burst. The same could happen if a loaded gun was accidentally reloaded.



EARLY BREECH-LOADING PAPER CARTRIDGE

- **GAS LEAKAGE** was a problem with early breech-loading guns, which used cartridges made of paper and other combustible material. Leakage reduced the pressure of the exploding gas that propelled the projectile.

